EE5327 VLSI Design Project, Report 1

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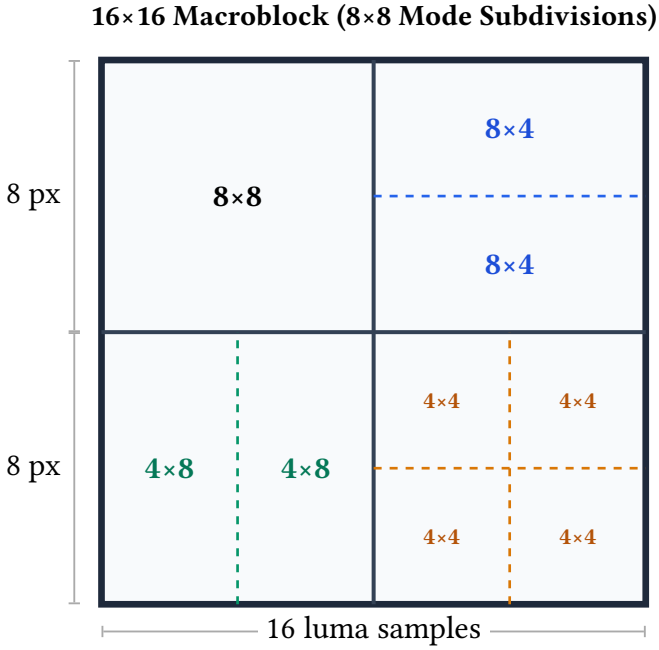
Introduction

As discussed in [1], motion estimation is the most computationally intensive task in H.264/H.265 video encoding, accounting for around 80% of the total computational for video encoding. As the name implies, motion estimators improve encoding efficiency by identifying movement over time in the source video. The objective of any motion estimation algorithm is to produce an array of optimal motion vectors (MVs) that describe change over time between the current frame and a previous reference frame. Compression standards like Advanced Video Coding (AVC/H.264) and High Efficiency Video Coding (HEVC/H.265) support variable-block-size prediction. In H.264, a variable-block-size motion estimation algorithm produces the optimal displacement vector for each 16x16 macroblock and all of their supported subdivisions (4x4, 4x8, 8x4, 8x8, 8x16, and 16x8).



Formally,

Various architectures and algorithms have been proposed for efficient motion estimation. In software, advanced fast-search algorithms like Enhanced Predictive Zonal Search (EPZS) and variants of the cross-diamond-hexagonal search

In hardware, algorithms like three step search (TSS), four step search (4SS) and diamond search (DS), can accelerate

Bibliography

[1] N. Cheeckottu Vayalil and Y. Kong, "VLSI Architecture of Full-Search Variable-Block-Size Motion Estimation for HEVC Video Encoding," *IET Circuits, Devices & Systems*, vol. 11, no. 6, pp. 543–548, 2017, doi: https://doi.org/10.1049/iet-cds.2016.0267.